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AlSiMgX MASTER ALLOY AND USE OF THE MASTER ALLOY IN THE PRODUCTION OF AN ALUMINIUM ALLOY

Abstract

The invention relates to an AlSiMgX master alloy, especially suitable for increasing the Magnesium concentration of aluminium foundry alloy melts. The invention also relates to a process for increasing the Mg content of aluminium foundry alloys by use of the AlSiMgX master alloy in preparing a target aluminium alloy. The AlSiMgX master alloy comprises: Mg 1.3-6.5 wt %, Si 6.5-11.5 wt %, Cu 0-1 wt %, Mn 0-1.0 wt %, Fe $\leq$ 0.40 wt %, Ti $\leq$ 0.18 wt %, Sr $\leq$ 0.10 wt %, balance Al and incidental impurities.

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Background/Summary

TECHNICAL FIELD

[0001] The invention relates to a master alloy for increasing the Magnesium content of aluminium alloy melts, especially aluminium foundry alloy melts. The invention also relates to a process for preparing a target aluminium alloy by increasing the Mg content of aluminium foundry alloys, and the use of the master alloy for increasing the Mg content of aluminium foundry alloys.

BACKGROUND

[0002] Most of the aluminium foundry alloys used for shape-casting are based on Si and Mg as major alloying elements. Si gives an alloy good castability, and together with Mg it enables precipitation hardening. For gravity and low-pressure casting, the AlSi7Mg-type of alloys, listed e.g. as EN AC-AlSi7Mg0.3 in the European Standard EN1706 with a Mg-range of 0.25-0.45 wt %, is by far the most used primary foundry alloys worldwide, while variants with higher Mg content (up to 0.7 wt %) or with some copper addition (up to 1 wt %) are also commonly used. For high pressure die casting of structural parts, AlSiMnMg alloys with a Si content from 7 to 10 wt % and a Mg-content up to 0.6 wt % have become very popular.

[0003] All the aforementioned alloys are usually heat-treated to temper T5, T6 or T7 where the Mg-content plays a crucial role and allows to tailor-make the mechanical properties. This is illustrated in FIG. 1 showing the mechanical properties of AlSi7 alloys in T6 temper as a function of the Mg content. It is therefore important for the foundries to adjust the Mg-content on a specific level depending on the mechanical properties requirements.

[0004] Many foundries increase the Mg-content of their melts on a regular basis for one or both of the following reasons. Firstly, some cast products may require a higher strength than what is achievable with their commonly purchased alloy. Obtaining this by a specific increase of the Mg-content of the alloy eliminates the need for a second alloy. Secondly, most foundries remelt their internal scrap consisting of feeders, gating system, scrapped parts or chips from machining operations. During remelting of such scrap, often in shape of small pieces, Mg tends to burn-off to a certain extent leading to a decrease of the Mg-content in the melt. Mg additions to the melt are therefore needed to compensate for the Mg burn-off.

[0005] Increasing the Mg-content of aluminium alloys is traditionally done by adding pure Mg metal to the melt in a transport crucible before degassing or directly in the casting furnace. If not immersed by appropriate tools the pure Mg metal tends to float on the melt surface, which leads to enhanced burn-off and formation of oxides. Measurements have shown that this practice, besides a poor and unpredictable yield due to burn-off of Mg, very often leads to a significant increase in oxide content in the melt (Mg-oxides and mixed AlMg-oxides/spinel).

[0006] There is thus a desire for an improved method of increasing the content of Mg of aluminium melts that provides a solution to the challenges mentioned in the previous paragraphs.

SUMMARY OF THE INVENTION

[0007] The present invention provides a solution that alleviates at least some or all of the problems associated with the traditional methods for increasing the Mg content in aluminium alloys.

[0008] The present invention solves, or at least alleviates the problems associated with the traditional methods for increasing the Mg content in aluminium alloys, by providing an AlSiMgX master alloy for use in preparing aluminium-silicon based foundry alloys comprising Mg.

[0009] The present invention also provides a method for preparing an aluminium target alloy by use of said AlSiMgX master alloy.

[0010] In the present disclosure the following terms should be interpreted and understood according to the following definitions, unless otherwise stated:

[0011] As used herein, the term “base alloy” means an initial aluminium-based alloy prepared according to common methods in a melting furnace after melting of input material such as ingots, pre-alloyed ingots, master alloys, alloying elements, process scrap, feeders, chips etc.

[0012] As used herein, the term “master alloy” means an aluminium alloy rich in certain alloying elements to be added to a base alloy in order to achieve/prepare a target aluminium alloy composition. In the present disclosure the AlSiMgX master alloy is rich in Mg, while the other alloying elements essentially corresponds with the aluminium base alloy, and hence the aluminium target alloy.

[0013] As used herein, the term “target alloy” means an aluminium alloy achieved after adding the alloying elements, e.g. by using a master alloy, to the base aluminium alloy to achieve a target composition of the aluminium alloy. In the present disclosure the term “target aluminium alloy” should be understood to denote an aluminium base alloy to which the AlSiMgX master alloy has been added resulting in an increase of the Mg concentration in the target aluminium alloy, compared with the aluminium base alloy

[0014] As used herein, the term “alloying element” means any purposeful addition of an element to a base metal, in this disclosure aluminium, for the purpose of improving its processability or modifying the mechanical, corrosion, electrical or thermal characteristics through modification of the metallurgical structure of the base metal. The term does not include incidental impurities, unless expressively stated differently.

[0015] The present invention provides, according to a first aspect, an AlSiMgX master alloy for increasing the content of Mg in an aluminium base alloy in the preparation of a target aluminium alloy (the term “content of Mg” may also be referred to as “concentration of Mg” herein). The aluminium base alloy may be an AlSiMg alloy. It is preferred that the aluminium base alloy is an AlSiMg alloy having a composition essentially corresponding to the AlSiMgX master alloy, except for the Mg concentration which is generally lower in the aluminium base alloy compared with the AlSiMgX master alloy. The target aluminium alloy may be an AlSiMg alloy. It is preferred that the aluminium target alloy is an AlSiMg alloy having a composition essentially corresponding to the AlSiMgX master alloy and the aluminium base alloy, except for the Mg concentration which is generally lower compared with the AlSiMgX master alloy but higher compared with the aluminium base alloy.

[0016] The AlSiMgX master alloy, according to the first aspect, has the following composition:

[0017] Mg 1.3-6.5 wt %, [0018] Si 6.5-11.5 wt %, [0019] Cu 0-1.0 wt %, [0020] Mn 0-1.0 wt %, [0021] Fe≤0.40 wt %, [0022] Ti≤0.18 wt %, [0023] Sr≤0.10 wt %, and [0024] the rest being Al and incidental impurities.

[0025] The respective concentrations of the alloying elements selected from the group comprising Si, Cu, Mn, Fe, Ti and Sr in the AlSiMgX master alloy should correspond to the concentrations of the same alloying elements in the aluminium base alloy to which the AlSiMgX master alloy is to be added, while the Mg content in the AlSiMgX master alloy can be significantly higher than the said aluminium base alloy. By keeping the concentrations of the said alloying elements (except for Mg) in the AlSiMgX master alloy corresponding to the concentrations of the said alloying elements in the base alloy it is possible to adjust the Mg content without altering the overall chemistry of the base alloy/target alloy. The present AlSiMgX master alloy gives a predictable yield in the aluminium target alloy prepared by usage of the AlSiMgX master alloy, it does not deteriorate the melt quality, and it does not change the overall chemical composition by diluting effects. Any burn-off of Mg can be adjusted in a reliable and consistent way by addition of an appropriate amount of an AlSiMgX master alloy having an appropriate base composition.

[0026] The content of Mg in the AlSiMgX master alloy is from 1.3 to 6.5 wt %. In alternative embodiments the content of Mg in the AlSiMgX master alloy may suitably be from 1.3 to 5.5 wt %; such as from 1.3 to 2.5 wt %; 1.5 to 2.5 wt %; 1.5 to 2.0 wt %; 2.0 to 3.0 wt %; 2.5 to 3.5 wt %; 3.4 to 4.0 wt %; 4.0 to 5.5 wt %; 4.5 to 5.5 wt %, or from 4.0 to 4.6 wt %. The alternative concentration ranges of Mg in the AlSiMgX master alloy allows preparation of different AlSiMgX master alloy compositions which may be tailored for desired increase of Mg content in different target aluminium alloys, compared with the base aluminium alloy.

[0027] The content of Si in the AlSiMgX master alloy is between 6.5 and 11.5 wt %. The content of Si in the AlSiMgX master alloy may be from 6.5 to 7.5 wt %, or from 9 to 11.5 wt %. The said Si ranges are used in many aluminium foundry alloys based on Si and Mg as major alloying elements. Therefore, AlSiMgX master alloys comprising Si in the said ranges are especially suitable for adjusting the Mg content of such AlSiMg foundry alloys. In an example, the content of Si in the AlSiMgX master alloy is from 6.5 to 11.5 wt %, and the content of Mn is from 0.4 to 0.8 wt %. In another example, the content of Si in the AlSiMgX master alloy is from 6.5 to 8.5 wt %, and the content of Mn is from 0.4 to 0.8 wt %. In a further example, the content of Si in the AlSiMgX master alloy is from 9.0 to 11.5 wt %, and the content of Mn is from 0.4 to 0.8 wt %. The AlSiMgX master alloys comprising Si and Mn are especially suitable for adjusting the Mg content of AlSiMnMg foundry alloys.

[0028] In an example of the AlSiMgX master alloy, the content of Si is from 6.5 to 7.5 wt % and the content of Cu is from 0.2 to 0.7 wt %. The AlSiMgX master alloys comprising Si and Cu are especially suitable for adjusting the Mg content of AlSiCuMg foundry alloys.

[0029] The content of Fe in the AlSiMgX master alloy is up to 0.40 wt %. In an alternative embodiment, the amount of Fe in the AlSiMgX master alloy may be 0.15 wt % or less. The content of Ti in the AlSiMgX master alloy is up to 0.18 wt %, In an alternative embodiment, the amount of Ti in the AlSiMgX master alloy may be from 0.05 to 0.15 wt %. The content of Sr in the AlSiMgX master alloy is up to 0.10 wt %. In an alternative embodiment, the amount of Sr in the AlSiMgX master alloy may be from 0.02 to 0.04 wt %.

[0030] The AlSiMgX master alloy may be based on 3xx alloys (according to the nomenclature of the Aluminium Association designation system), EN AC-42xxx alloys (AlSi7Mg), EN AC-43500 alloys (AlSi10MnMg), or EN AC-45500 alloys (AlSi7Cu0.5 Mg) (according to European Standard EN1706 and/or EN 1676), wherein the AlSiMgX master alloy has a higher Mg concentration compared to the said standard alloys, the Mg content being within 1.3 to 6.5 wt %, such as 1.3 to 5.5 wt %, or 1.3 to 2.5 wt %, or 1.5 to 2.5 wt %, or 1.5 to 2.0 wt %, or 2.0 to 3.0 wt %, or 2.5 to 3.5 wt %, or 3.4 to 4.0 wt %, or 4.0 to 5.5 wt %, or 4.5 to 5.5 wt %, or 4.0 to 4.6 wt %. The said standard casting alloys represent common foundry alloys. AlSiMgX master alloys based on such standard casting alloys but having an increased amount of Mg compared with the standard composition, can be tailor-made for the standard foundry alloys and for specific increase of the concentration of Mg without changing the overall chemistry of the alloy.

[0031] The AlSiMgX master alloy may be in the form of an ingot, rod, wire, pellet, briquet, foil, waffle, button, rod, powder or splatter. Preferably the AlSiMgX master alloy is in the form of an ingot. An AlSiMgX master alloy ingot may have a weight of 6.5, 7.5 or 9.2 kg, or other standard ingot weight used in the aluminium smelting industry.

[0032] According to a second aspect of the invention, the present disclosure provides a method for adjusting the concentration of Mg in an aluminium base alloy in the preparation of an aluminium target alloy, the method comprises [0033] providing an AlSiMgX master alloy according to the present disclosure and the first aspect of the present disclosure, [0034] providing an aluminium base alloy having essentially the same composition as the AlSiMgX master alloy except the Mg concentration being lower compared to the Mg concentration of the AlSiMgX master alloy, [0035] adding a predetermined amount of the AlSiMgX master alloy to the aluminium base alloy, thereby increasing the Mg content of the aluminium base alloy while keeping the concentration of any

other alloying elements essentially unchanged.

[0036] By the present method the concentration of any other alloying elements chosen from the group comprising Si, Cu, Mn, Fe, Ti, Sr in the aluminium target alloy are essentially unchanged compared with the aluminium base alloy after addition of the AlSiMgX master alloy. The term “essentially” should in this context and in the present disclosure be understood as the concentration of the said alloying elements are not changed in an amount that would alter the characteristics of the aluminium target alloy, as the modification of the aluminium target alloy is based on the adjustment of the Mg concentration.

[0037] The AlSiMgX master alloy may be added to a melting furnace, prior to, during, or after melting the aluminium base alloy.

[0038] The AlSiMgX master alloy may be added to a transport crucible, prior to, during, or after being filled with molten aluminium base alloy.

[0039] The AlSiMgX master alloy may be added to a holding or casting furnace, prior to, during, or after pouring in aluminium base alloy.

[0040] The AlSiMgX master alloy may be added to a continuously melting furnace, wherein the AlSiMgX master alloy may be periodically fed to the furnace to compensate for Mg-loss.

[0041] The AlSiMgX master alloy may be added in the form of an ingot, rod, wire, pellet, briquet, foil, waffle, button, rod, powder or splatter.

[0042] According to a third aspect, the present disclosure provides a method for preparing the AlSiMgX master alloy according to the present disclosure, the method comprising: providing an aluminium base alloy in solid or molten state; if necessary determining the concentration, in weight percent, of each alloying element in the aluminium base alloy; estimating and adding an appropriate amount of Mg into the aluminium base alloy to obtain a desired concentration of Mg in the AlSiMgX master alloy, and casting the molten AlSiMgX master alloy. The aluminium base alloy, on which the AlSiMgX master alloy is based, may be chosen from a 3xx alloy (according to the nomenclature of the Aluminium Association designation system), EN AC-42xxx, EN AC-43500 (AlSi10MnMg) or EN AC-45500 (AlSi7Cu0.5 Mg) alloy (according to European Standard EN1706 and/or EN 1676).

[0043] The method for preparing the AlSiMgX master alloy may comprise adjusting the amount of any other alloying elements selected from the group consisting of Si, Cu, Fe, Mn, Ti and Sr by adding an appropriate amount of the alloying element(s) to the aluminium base alloy to obtain the desired composition of the AlSiMgX master alloy. Preferably the method comprises casting the AlSiMgX master alloy into an ingot, rod, wire, pellet, or briquet. The cast AlSiMgX master alloy may be further processed to be in the form of a foil, waffle, button, rod, wire, powder or splatter.

[0044] According to a fourth aspect, the present disclosure further provides the use of the AlSiMgX master alloy according to the present disclosure, wherein the AlSiMgX master alloy is added to an aluminium base alloy to increase the content of Mg in the aluminium base alloy while keeping the concentration of any other alloying elements in the aluminium base alloy essentially unchanged.

Description

DETAILED DESCRIPTION OF THE DRAWINGS

[0045] FIG. 1 illustrates the mechanical properties of AlSi7 in T6 temper as a function of the Mg-content.

DETAILED DESCRIPTION OF THE INVENTION

[0046] The present disclosure relates to an AlSiMgX master alloy especially suitable for increasing the Mg content of aluminium foundry alloy melts. The AlSiMgX master alloys according to the present disclosure is therefore especially suitable for usage in a process for preparing target aluminium alloys. Preferably, the AlSiMgX master alloy contains overall the same alloying

elements and essentially in the same concentrations that are desired in the target aluminium alloy, except for a higher content of Mg.

[0047] The AlSiMgX master alloy according to the present disclosure comprises: [0048] Mg 1.3-6.5 wt %; [0049] Si 6.5-11.5 wt %; [0050] Cu 0-1.0 wt %; [0051] Mn 0-1.0 wt %; [0052] Fe $\leq$ 0.40 wt %; [0053] Ti $\leq$ 0.18 wt %; [0054] Sr $\leq$ 0.10 wt %; the rest being Al and incidental impurities.

[0055] The AlSiMgX master alloy is used in a process for preparing a target aluminium alloy by addition of an appropriate amount of the above AlSiMgX master alloy to an aluminium base alloy.

[0056] The composition of any particular AlSiMgX master alloy according to the present disclosure depends upon the composition of the desired target aluminium alloy, which also corresponds with the composition of the base aluminium alloy. Except for the concentration of Mg, the AlSiMgX master alloy shall have essentially the same concentrations of alloying elements (alloying elements selected from the group consisting of Si, Cu, Fe, Mn, Ti and Sr) as the target alloy. By “essentially the same concentrations” is meant that the concentrations of the said alloying elements (except Mg) in the AlSiMgX master alloy corresponds with the target aluminium alloy, such that the resulting concentration of the said alloying elements in the target aluminium alloy, after addition of the AlSiMgX master alloy to a base aluminium alloy, have generally corresponding concentration except the concentration of Mg which is increased in the target aluminium alloy compared to the base alloy. For a given aluminium base alloy, an AlSiMgX master alloy of the invention can be prepared. It should be appreciated that the weight percent concentration of each alloying element in the base alloy can be identified by techniques, generally known in the art.

[0057] The AlSiMgX master alloy may be used in a process for preparing a target aluminium alloy by addition of the AlSiMgX master alloy to an aluminium base alloy. A preferred aluminium base alloy may be selected from the 3xx series, as designated by the Aluminium Association or from other European or national foundry alloys. The aluminium base alloy may be an EN AC-42xxx, EN AC-43500 (AlSi10MnMg) or EN AC-45500 (AlSi7Cu0.5 Mg) alloy, according to European Standard EN1706 and/or EN1676. The base alloy, to which the AlSiMgX master alloy is added, determines the alloying elements and their concentration in the AlSiMgX master alloy.

[0058] The Mg concentration in the AlSiMgX master alloy is from 1.3 to 6.5 wt %. In some applications, the Mg concentration in the AlSiMgX master alloy may range from 1.3 to 5.5 wt %, or 1.3 to 4.5, or 1.3 to 2.5 wt %, or 1.5 to 2.5 wt %, or 1.5 to 2.0 wt %, or 2.0 to 3.0 wt %, or 2.5 to 3.5 wt %, or 3.4 to 4.0 wt %, or 4.0 to 5.5 wt %, or 4.5 to 5.5 wt %, or 4.0 to 4.6 wt %. The concentration of Mg in the AlSiMgX master alloy generally depends on the desired increase of Mg in the target aluminium alloy, compared with the base aluminium alloy.

[0059] The Si concentration in the AlSiMgX master alloy is from 6.5 to 11.5 wt %. In some examples, the Si concentration in the AlSiMgX master alloy may range from 6.5 to 7.5 wt %. Some gravity and low-pressure aluminium casting alloys are of the AlSi7Mg-type, hence AlSiMgX master alloys having between 6.5-7.5 wt % Si are especially suited for such AlSi7Mg-type alloys. In another example, the Si concentration in the AlSiMgX master alloy may range from 9 to 11.5 wt % which are particularly suitable for addition to AlSi10Mg-type alloys. An AlSiMgX master alloy comprising from 6.5 to 11.5 wt % Si, e.g. from 6.5 to 8.5 wt % Si, or from 9.0 to 11.5 wt % Si, together with 0.4 to 0.8 wt % Mn are especially suitable for additions to AlSiMnMg-type alloys, such as AlSi7MnMg-type and AlSi10MnMg-type alloys. Furthermore, an AlSiMgX master alloy comprising from 6.5 to 7.5 wt % Si and 0.2 to 0.7 wt % Cu are especially suitable for addition to AlSi7MgCu0.5 type alloys.

[0060] The Fe concentration in the AlSiMgX master alloy should be 0.40 wt % or less, such as 0.15 wt % or less.

[0061] The Ti concentration in the AlSiMgX master alloy should be 0.18 wt % or less, such as between 0.05 and 0.15 wt %.

[0062] The Sr concentration in the AlSiMgX master alloy should be 0.10 wt % or less, such as

between 0.02-0.04 wt %.

[0063] It should be appreciated that the composition of the AlSiMgX master alloy may be varied by combining the exemplified ranges of the alloying elements, and within the generally defined composition according to the appended claims.

[0064] The AlSiMgX master alloys of the present disclosure are prepared according to generally known techniques of production of foundry alloys. Commercially pure aluminium, scrap aluminium alloy, including recycled aluminium metal, or a combination thereof, as well as the alloying elements of the respective base alloy can be used as the starting material. The AlSiMgX master alloy is preferably based on a 3xx series alloy (according to the nomenclature of the Aluminium Association designation system), EN AC-42xxx, EN AC-43500 (AlSi10MnMg) or EN AC-45500 (AlSi7Cu0.5 Mg) alloy (according to European Standard EN1706 and/or EN 1676), having an increased Mg concentration compared with the said standard base alloys. A sufficient amount of magnesium is used to provide the calculated final concentration of magnesium in the AlSiMgX master alloy. After the final element has been added, it is desirable to immediately adjust the temperature so as to provide fluidity for casting and, depending on furnace stirring characteristics, provide a product that, when cast, is of consistent chemistry from the beginning to the end of the charge so as to remove concerns about segregation.

[0065] The casted AlSiMgX master alloy may be further processed or the final step in its preparation may be modified so as to produce the AlSiMgX master alloy in any desirable form. Such forms include foil, waffle, ingot, button, rod, wire, pellet, powder, briquet, and splatter. However, in many applications the preferred form of the AlSiMgX master alloy is an ingot. The ingot can be in a corresponding shape and/or size of the ingots used for the base alloy ingots. Another preferred form of the AlSiMgX master alloy is a continuous cast (direct chill) ingot, where the developing Mg-containing phases are small in size and well distributed due to the high solidification rate. An AlSiMgX master alloy ingot wherein the Mg-containing phases are small in size and well distributed facilitates a rapid dissolution of the master alloy when added to a molten aluminium base alloy.

[0066] The AlSiMgX master alloy according to the invention is used in the preparation of final target aluminium alloys. The AlSiMgX master alloy can be added to a melting furnace (prior to, during, or after melting), to the melt in a transport crucible or in a holding or casting furnace. For example, for a single furnace melting system, where the cast alloy is cast from the melting furnace, the base aluminium alloy is often prepared by using pre-alloyed foundry aluminium alloy ingots. Alternatively, the base aluminium alloy may be prepared, using commercially pure aluminium, scrap aluminium alloy, including recycled aluminium metal, or a combination thereof as well as the required alloying elements. Sufficient material is added until the basic charge weight is achieved, except for the amount of the AlSiMgX master alloy which may be added subsequently. The temperature is raised to above the melting point, typically between 70° and 800° C. Then an appropriate amount of AlSiMgX master alloy material may be added to achieve the desired final chemistry of the target alloy. Advantageously, the surface of the molten aluminium base alloy is skimmed clean of any oxides before the AlSiMgX master alloy addition.

[0067] Alternatively, depending on the expected nominal Mg-content calculated based on the composition of the input material (base aluminium alloy) an appropriate amount of AlSiMgX master alloy can also be added to the input material prior to melting in order to directly achieve the desired Mg-content after complete melting of the input and the AlSiMgX master alloy materials.

[0068] In another example, the AlSiMgX master alloys according to the present disclosure can be added to continuously melting furnaces such as shaft furnaces or chip-melting furnaces, where the normal expected Mg-loss due to burn-off during melting is usually quite well known. Depending on the hourly melting capacity and the necessary compensation of the Mg-loss or desired increase of the Mg-content, AlSiMgX master alloy ingots can be fed into the melt, e.g. in regular periods.

[0069] In a further example, the AlSiMgX master alloys according to the present disclosure can be

added to the base aluminium alloy in a holding or casting furnace as the base alloy metal is being poured in. This provides a stirring action and minimizes the time and temperature for making alloying additions, thereby minimizing oxidation of some alloying elements, especially Mg. An alternative where the AlSiMgX master alloy is added to a filled holding furnace appropriate stirring would be needed.

[0070] In still another alternative, the AlSiMgX master alloys according to the present disclosure can be added outside of the melting furnace, such as in a transport crucible, so that the base alloy melt composition in the melting furnace is not affected. Preferably the addition of the AlSiMgX master alloy takes place into the empty transport crucible before it is being filled or to the melt in a transport crucible prior to an eventual rotor degassing process. Again, this provides stirring action and minimizes the time and temperature for making alloying additions, thereby minimizing oxidation of some alloying elements.

[0071] In a preferred embodiment of the invention the AlSiMgX master alloy is in the form of ingots having a size and a Mg-concentration that results in a stepwise and/or fixed increase of the Mg concentration when one or more AlSiMgX master alloy ingots are added to a certain amount of base aluminium alloy. The AlSiMgX master alloy ingots may have a weight of 6.5 kg, 7.5 kg or 9.2 kg, which sizes corresponds to commonly used base aluminium ingots. Addition of one AlSiMgX master alloy ingot to a certain amount of the base aluminium alloy may increase the Mg concentration in the target aluminium alloy by e.g. 0.1 wt % (percentage point). Table 1 shows examples of AlSiMgX master alloy ingots sizes and Mg ranges resulting in a fixed Mg increase (in wt %) in target aluminium alloys by addition of one ingot to 1000 kg base alloy. Using such "standardized" AlSiMgX master alloy ingot system facilitates and simplifies the production of target aluminium alloys. It should be noted that Table 1 only illustrates example alloys for better understanding the invention. The examples illustrated in Table 1 should therefore not be construed as limiting for the present invention since there are a variety of possible master alloy compositions within the defined claim scope in the appended claims.

TABLE-US-00001 TABLE 1 Desired Mg increase in target alloy by adding 1 Mg range in Ingot weight ingot to 1000 kg of melt master alloy [kg] [wt %] [wt %]

6.5	0.01	1.7-2.1	6.5	0.02	3.4-3.8
6.5	0.03	4.8-5.2	7.5	0.01	1.5-1.9
7.5	0.02	2.9-3.3	7.5	0.03	4.2-4.6
9.2	0.01	1.3-1.7	9.2	0.02	2.5-2.9
9.2	0.03	3.5-3.9			

[0072] The following examples illustrate the use and advantages of the AlSiMgX master alloys according to the present disclosure. The illustrating examples should not be construed as limiting for the present invention.

[0073] The first illustrating example concerns a 3xx base alloy containing nominally 7 wt % Si, 0.12 wt % Fe, 0.12 wt % Ti, 0.02 wt % Sr, and 0.28 wt % magnesium, balance Al. In order to fulfil required strength requirements, the Mg concentration shall be increased to 0.29 wt %. By adding an AlSiMgX master alloy ingot of 6.5 kg having the same nominal composition as the base alloy, but with 1.9 wt % Mg to 1000 kg of the above-described base alloy melt will nominally increase the Mg-content of the melt by 0.01 wt %, i.e. to a final Mg content of 0.29 wt %. In the same way, the addition of two ingots would nominally increase the Mg-content by 0.02 wt %. A calculated nominal loss in melt temperature of approx. 7.5 degrees per added ingot to a 1000-kg-melt has to be taken into consideration. The time for melting and complete dissolution of the AlSiMgX master alloy ingot can be expected to be within 2-3 minutes. Pouring the liquid base alloy onto the master alloy ingot or stirring (e.g. by a rotor degassing treatment) would further decrease the melting and dissolution time.

[0074] The second illustrating example concerns a 3xx base alloy containing nominally 7 wt % Si, 0.12 wt % Fe, 0.12 wt % Ti, 0.02 wt % Sr, and 0.28 wt % magnesium, balance Al. In order to fulfil required strength requirements, the Mg concentration shall be increased to 0.32 wt %. By adding two AlSiMgX master alloy ingots of 6.5 kg having the same composition as the base alloy, but with 3.8 wt % Mg to 1000 kg of the above-described base alloy melt will nominally increase the Mg-

content of the melt by 0.04 wt % i.e. to a final value of 0.32 wt %.

[0075] The AlSiMgX master alloys of the present invention provide several advantages over conventional methods of adding Mg to aluminium alloys. The AlSiMgX master alloys provide a concentrated amount of Mg in the proper proportion that is required to produce the specific target alloy, thereby allowing the desired composition to be reached with the addition of only one master alloy. Due to finely distributed Mg-phases the AlSiMgX master alloys provide high solution rates, thereby reducing furnace cycle time or process time. The AlSiMgX master alloys reduce losses, e.g. caused by burn-off, and furthermore, they reduce melt treatment time, both due to faster dissolution and easier handling compared with the traditional methods. The AlSiMgX master alloys also provide, in certain instances, more consistent chemistry control due to more reliable yield compared to addition of pure Mg or binary AlMg alloys. These advantages result in increased efficiency and decreased manufacturing costs for shape-casting foundries.

[0076] The following examples relate to AlSiMgX master alloys according to the disclosure:

Example 1

[0077] 233.1 kg of an AlSi7Mg alloy was prepared in an electric resistance furnace. The temperature of the melt was 740° C. The composition of the alloy analyzed with Optical Emission Spectroscopy (OES) is given in table 2. The Mg concentration was 0.309 wt %.

[0078] An AlSiMgX master alloy according to the invention was added to the 233.1 kg melt. The composition of the AlSiMgX master alloy is given in table 2. The Mg concentration of the master alloy was 1.37 wt %. An amount of master alloy (5.009 kg) representing a theoretical increment in Mg concentration of 0.0223 wt %. was added to the melt. After dissolution of the master alloy the Mg concentration increased by 0.0249 wt %, giving an estimated yield of 111.5%. A person skilled in the art would conclude the deviation from 100% yield would be due to uncertainty in the chemical analyses (OES).

TABLE-US-00002 TABLE 2 Weight Si Mg Fe Ti Sr Others each Alloy [kg] [wt %] [wt %] [wt %] [wt %] [wt %] [wt %] AlSi7Mg 233.1 7.40 0.3091 0.1250 0.1143 0.0248 <0.01 (base alloy) Master alloy 5.009 7.50 1.37 0.1240 0.1031 0.0377 <0.01 AlSi7Mg 238.1 7.40 0.3340 0.1270 0.1149 0.0251 <0.01 (target alloy) Expected Mg increment 0.0223 in AlSi7Mg [wt %] Observed Mg increment 0.0249 in AlSi7Mg [wt %] Yield [%] 111.5

Example 2

[0079] 1.585 kg of an AlSi7Mg alloy was melted in an electric resistance furnace. The temperature of the melt was 720° C. The composition of the alloy analyzed with OES is given in table 3. The Mg concentration was 0.3043 wt %. An AlSiMgX master alloy according to the invention was added to the 1.585 kg melt. The composition of the master alloy is given in table 3. The Mg concentration of the master alloy was 1.37 wt %. An amount of master alloy (0.1592 kg) representing a theoretical increment in Mg concentration of 0.0973 wt %. was added to the melt. After dissolution of the master alloy the Mg concentration increased by 0.0957 wt %, giving an estimated yield of 98.4%. Again, the deviation from 100% yield can be explained by OES analysis uncertainty.

TABLE-US-00003 TABLE 3 Weight Si Mg Fe Ti Sr Others each Alloy [kg] [wt %] [wt %] [wt %] [wt %] [wt %] [wt %] AlSi7Mg 1.5850 6.96 0.3043 0.0989 0.1343 0.0251 <0.01 (base alloy) Master alloy 0.1592 7.50 1.37 0.1240 0.1031 0.0377 <0.01 AlSi7Mg 1.7442 6.98 0.4000 0.1014 0.1392 0.0250 <0.01 (target alloy) Expected Mg increment 0.0973 in AlSi7Mg [wt %] Observed Mg increment 0.0957 in AlSi7Mg [wt %] Yield [%] 98.4

Example 3

[0080] In a wheel foundry, a transport crucible was filled with 891 kg of an AlSi7Mg melt from a melting furnace. The weight of the melt was determined by weighing the crucible before and after filling. The composition of the alloy analyzed with Optical Emission Spectroscopy (OES) is given in table 4. The Mg concentration was 0.3021%. An AlSiMgX master alloy according to the invention was added to the 981 kg melt. The composition of the AlSiMgX master alloy is given in

table 4. The Mg concentration of the master alloy was 1.37 wt %. One ingot of the master alloy (9.2 kg) representing a theoretical increment in Mg concentration of 0.011 wt % was added to the melt prior to a degassing treatment of four minutes. In an OES sample taken just after the treatment, the Mg concentration increased to 0.3142 wt %, i.e. by 0.0121 wt %, giving an estimated yield of 111%. A person skilled in the art would conclude the deviation from 100% yield would be due to uncertainty in the chemical analyses (OES).

TABLE-US-00004 TABLE 4 Weight Si Mg Fe Ti Sr Others each Alloy [kg] [wt %] [wt %] [wt %] [wt %] [wt %] [wt %] AlSi7Mg 891 7.4658 0.3021 0.1126 0.1203 0.0211 (base alloy) Master alloy 9.2 7.4950 1.37 0.1240 0.1031 0.0377 <0.01 Upgraded 900.2 7.4952 0.3142 0.1153 0.1232 0.0204 AlSi7Mg (target alloy) Expected Mg increment 0.0109 in AlSi7Mg [wt %] Observed Mg increment 0.0121 in AlSi7Mg [wt %] Yield [%] 111.0

[0081] The three examples show that the composition of the target alloy has essentially the same composition of the alloying elements as the base alloys, except the amount of Mg which is increased to the desired concentration.

[0082] The specific details described in the context of the various example embodiments are not intended to be construed as limitations. The disclosed examples and alternatives of the invention may be readily combined, without departing from the scope as defined in the appended claims.

Claims

1. An AlSiMgX master alloy for increasing the content of Mg in an aluminium base alloy in the preparation of a target aluminium alloy, the AlSiMgX master alloy comprises: Mg 1.3-6.5 wt %; Si 6.5-11.5 wt %; Cu 0-1.0 wt %; Mn 0-1.0 wt %; Fe≤0.40 wt %; Ti≤0.18 wt %; Sr≤0.10 wt %; and balance Al and incidental impurities.
2. The AlSiMgX master alloy according to claim 1, wherein the content of Si is 6.5-7.5 wt %, or 9.0-11.5 wt %.
3. The AlSiMgX master alloy according to claim 1, wherein the content of Si is 6.5-11.5 wt % and the content of Mn is 0.4-0.8 wt %.
4. The AlSiMgX master alloy according to claim 3, wherein the content of Si is 6.5-8.5 wt %, or 9.0-11.5, and the content of Mn is 0.4-0.8 wt %.
5. The AlSiMgX master alloy according to claim 1, wherein the content of Si is 6.5-7.5 wt % and the content of Cu is 0.2-0.7 wt %.
6. The AlSiMgX master alloy according to claim 1, wherein the content of Sr is 0.02-0.04 wt %.
7. The AlSiMgX master alloy according to claim 1, wherein the content of Fe is ≤0.15 wt %.
8. The AlSiMgX master alloy according to claim 1, wherein the content of Ti is 0.05-0.15 wt %.
9. The AlSiMgX master alloy according to claim 1, wherein the Mg content is 1.3-5.5 wt %; 1.3-2.5 wt %; 1.5-2.5 wt %; 1.5-2.0 wt %; 2.0-3.0 wt %; 2.5-3.5 wt %; 3.4-4.0 wt %; 4.0-5.5 wt %; 4.5-5.5 wt %, or 4.0-4.6 wt %.
10. The AlSiMgX master alloy according to claim 1, wherein the master alloy is based on 3xx alloy (according to the Aluminium Association designation system), EN AC-42xxx, EN AC-43500 (AlSi10MnMg) or EN AC-45500 (AlSi7Cu0.5 Mg) alloy (according to European Standard EN1706 and/or EN 1676), having an increased Mg concentration compared to the said standard alloys.
11. The AlSiMgX master alloy according to claim 1, where the AlSiMgX master alloy is in the form of an ingot, rod, wire, pellet, or briquet, foil, waffle, button, rod, powder or splatter.
12. (canceled)
13. A method for increasing the content of Mg in an aluminium base alloy in the preparation of a target aluminium alloy, the method comprises providing an AlSiMgX master alloy according to claim 1; providing an aluminium base alloy having essentially the same composition as the AlSiMgX master alloy except the Mg concentration being lower compared to the Mg concentration

- of the AlSiMgX master alloy; adding a predetermined amount of the AlSiMgX master alloy to the aluminium base alloy, thereby increasing the Mg content of the aluminium base alloy while keeping the concentration of any other alloying elements essentially unchanged.
- 14.** The method according to claim 13, comprising adding the AlSiMgX master alloy to a melting furnace, prior to, during, or after melting the aluminium base alloy.
- 15.** The method according to claim 13, comprising adding the AlSiMgX master alloy to a transport crucible, prior to, during, or after being filled with molten aluminium base alloy.
- 16.** The method according to claim 13, comprising adding the AlSiMgX master alloy to a holding or casting furnace, prior to, during, or after the aluminium base alloy is being poured in.
- 17.** The method according to claim 13, comprising adding the AlSiMgX master alloy to a continuously melting furnace, wherein the AlSiMgX master alloy is periodically fed to the furnace to compensate Mg-loss.
- 18.** The method according to claim 13, wherein the AlSiMgX master alloy is added in the form of an ingot, rod, wire, pellet, or briquet, foil, waffle, button, rod, powder or splatter.
- 19.** A method for preparing an AlSiMgX master alloy according to claim 1, the method comprises the steps: providing an aluminium base alloy in solid or molten state; optionally, determining the concentration, in weight percent, of each alloying element in the aluminium base alloy; adding an appropriate amount of Mg into the aluminium base alloy to obtain a desired concentration of Mg in the AlSiMgX master alloy, optionally, adjusting the amount of any other alloying elements selected from the group consisting of Si, Cu, Fe, Mn, Ti and Sr by adding an appropriate amount of the alloying element(s) to the aluminium base alloy to obtain the desired composition of the AlSiMgX master alloy, if necessary, adjusting temperature to provide fluidity for casting, and casting the AlSiMgX master alloy into an ingot, rod, wire, pellet, or briquet.
- 20.** The method according to claim 19, where the aluminium base alloy is chosen from a 3xx series alloy (according to the nomenclature of the Aluminium Association designation system); EN AC-42xxx alloy; EN AC-43500 alloy (AlSi10MnMg) or EN AC-45500 alloy (AlSi7Cu0.5 Mg) (according to European Standard EN1706 and/or EN 1676).
- 21-22.** (canceled)
- 23.** Use of an AlSiMgX master alloy according to claim 1, wherein a predetermined amount of the AlSiMgX master alloy is added to an aluminium base alloy to increase the content of Mg, while the concentration of any other alloying elements in the aluminium base alloy are essentially unchanged.
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